

THE SCIENCE OF MEASURING PEOPLE

What does a personality test have in common with a national census?

A lot more than you might think.

At first, the answer seems obvious: basically nothing.

A personality test asks questions like:

I enjoy trying new things.

You choose a number from 1 — Strongly Disagree to 5 — Strongly Agree.

A census survey asks questions like:

How many people live in your household?

How much money did you earn last year?

Does your home have running water?

One feels like psychology. The other feels like paperwork.

But underneath the surface, both are attempting something surprisingly difficult:

They are trying to turn human life into data.

And that turns out to be a much stranger problem than it sounds.

THE PROBLEM WITH MEASURING A HUMAN BEING

Let's start with something easy.

You can measure someone's height.

Put the person next to a measuring tape. Read the number. Done.

Now try measuring personality.

Where exactly do you put the measuring tape?

There isn't a physical object called "conscientiousness" sitting somewhere inside your brain waiting to be measured. The same problem appears when researchers want to measure things like poverty, unemployment, or housing quality.

These things are real.

They just aren't directly measurable in the same way that height is.

So researchers have to build a measurement system.

Psychologists might use a series of carefully designed questions to estimate personality traits. Statistical agencies might use income, household size, employment status, and other variables to determine whether a household falls below a particular poverty threshold.

This process has a name:

operationalization.

It's basically the scientific version of saying:

“Okay, we agree this thing exists. Now how are we actually going to measure it?”

That question matters enormously.

Change the measurement system, and you can change the result.

WELCOME TO THE WORLD OF CATEGORIES

Here's where things get interesting.

Human beings are messy.

Data hates mess.

Imagine asking one million people:

“Describe your financial situation.”

You would get one million different answers.

Somebody might say, “We're doing pretty well.”

Someone else might say, “We're okay, but groceries have gotten expensive.”

Someone else might say, “We're barely getting by.”

All of those answers contain useful information.

Good luck putting them into a spreadsheet.

So researchers create categories.

Income gets numbers. Employment gets definitions. Housing gets classifications. Personality questions get standardized response scales.

Suddenly, a gigantic collection of individual experiences becomes something researchers can analyze.

This is incredibly powerful.

It is also a little dangerous.

Because whenever you turn reality into categories, reality loses some of its detail.

A person who is “employed” could work 40 hours a week. Or two hours.

A household could technically have access to water while experiencing serious problems with its quality or reliability.

And a personality score can tell you something about a person without remotely capturing the whole person.

The categories are useful because they simplify reality.

The simplified version isn't reality itself.

WHO ANSWERED THE QUESTION?

Here's a detail about surveys that is easy to overlook.

Sometimes the person providing the information isn't the person the question is about.

Imagine a household survey.

A researcher asks one resident about everyone living in the home.

That person may know their partner's age. They probably know how many children live there.

But what about the exact income of another adult? Or details about someone's employment?

Now we've introduced a new source of uncertainty:

the informant.

Psychologists deal with something similar when one person rates another person's personality or behavior.

A parent might describe a child one way.

A teacher might describe the same child differently.

Neither person necessarily has the “correct” version. They are observing the same human being

from different places.

This creates a useful scientific question:

How much does the observer affect the observation?

That's a problem for psychologists.

It's also a problem for people counting millions of households.

THE BIGGEST SURVEY ISN'T NECESSARILY THE BEST SURVEY

Here's a thought experiment.

Suppose you want to know how much teenagers like reading.

You create a survey.

You get 100,000 responses.

Amazing.

Except there's a catch.

You advertised the survey exclusively in a huge online community for people who love books.

Now you have an enormous dataset.

You also have a pretty obvious problem.

The people who answered may be different from the people who didn't.

This is non-response bias.

And it matters because surveys are often used to make statements about an entire population.

If the people who disappear from your dataset are systematically different from the people who remain, your results can drift away from reality.

This is why statistical agencies use techniques such as survey weighting and adjustments to known population totals.

Think of weighting as a statistical attempt to say:

“Okay, these groups are showing up in our responses more often than we would expect. These other groups are showing up less. How can we account for that?”

The goal is to make the final dataset better reflect the population it is supposed to describe.

FROM FIVE PERSONALITY TRAITS TO MILLIONS OF PEOPLE

There is still a huge difference between personality research and a census.

The Big Five is interested in differences between individuals.

A national census or household survey is interested in patterns across a population.

The scale changes dramatically.

Imagine zooming out from one person until you can see an entire country.

At the individual level, you might ask:

How open to new experiences is this person?

At the population level, you might ask:

How many people live in this region?

Then:

How many are working?

Then:

How many have access to particular services?

The questions change because the scientific purpose changes.

But the underlying challenge remains surprisingly familiar.

Researchers have something complicated in front of them. They can't measure it directly.

So they create a system for turning pieces of observable information into evidence about the thing they actually care about.

That's the strange magic of measurement.

THE DATA ISN'T THE WORLD

This may be the most important idea here.

When you see a statistic saying that X% of people are employed, you're looking at the result of an enormous chain of decisions.

Someone had to decide what “employed” means.

Someone designed the questions.

Someone decided who should be surveyed.

Someone created categories for the answers.

Someone dealt with people who didn't respond.

Someone calculated the final statistic.

By the time the number appears in a report, it looks wonderfully simple.

That simplicity is the end product of a very complicated process.

And this is true far beyond censuses and personality tests.

Whenever we see a number describing human behavior, there is a story behind that number.

The interesting question isn't only:

“What does the data say?”

It's also:

“How did they turn human experience into this particular piece of data?”

Once you start asking that question, surveys become much more interesting.

And statistics become much harder to take for granted.

THINK LIKE A SCIENTIST

Imagine that you have been asked to measure quality of life across an entire country.

Where would you start?

You could measure income.

You could look at housing.

You could ask people about their health.

You could ask them how happy they are.

Each choice would tell you something.

Each choice would also leave something out.

How do you measure something as complicated as a human life?

That's the question sitting underneath almost every attempt to turn people into data.